

## Sylvia Micah & Shanty Lakapua

### Part A: Case Selection:

**Selected Case:** PNG Government payroll (Alesco). This case is chosen because of the repeated Auditor-General findings about significant overpayments, including to deceased officers, and the ongoing failure of controls within the national payroll system.

### Part B — Research Sources

To complete the analysis, the following two primary sources were identified and verified:

1. **Auditor-General's Performance Audit Report (Oct 2025):** *"Performance Audit Report on the effectiveness of PNG Public Service payroll management and expenditure 2016–2023"* by Gordon Kega. This document details the K439,039 offline payment anomaly and the lack of oversight in manual calculations.
  - **Source Location:** Published on the Office of the Auditor-General website and cited in *The National* (Jan 2026).
2. **News Analysis (Jan 2026):** *"Secretary confirms payment"* and *"Audit reveals government officer overpaid"* articles from The National newspaper. These articles provide the timeline of the audit release, the Department Secretary's response, and the specific context of the Alesco Version 12 limitations.
  - **Source Location:** The National online archives (cached on Moodle/accessible via campus network).

### Part C:

#### 1. What was the system meant to do, and for whom?

The Alesco system was designed to automate payroll and human resource management for over 90,000 Papua New Guinea public servants, ensuring that all payments are processed automatically and accurately, allowing for on-time payments and self-service access, which removed the need to use payslips. The Department of Personnel Management (data control), and the public servants relying on it for their livelihoods.

#### 2. What actually happened? Timeline of key events with your sources.

The Alesco payroll system was implemented between **2002–2004** (upgrading to Version 12). Since then, the system has remained stagnant for two decades, which forces the agencies to use manual (offline) calculations for benefits by **2023** due to system limitations (*The National*, 2014; Auditor-General Report, 2025). In **October 2025**, Auditor-General Gordon Kega revealed a **K439,039** payment to a deceased officer—four times their annual salary processed offline without verification (*The National*, Jan 2026). Despite a **2023** parliamentary recommendation to upgrade the software to prevent fraud, no action was taken, allowing these control failures to persist through **2025** (*PNG Business News*, 2023).

### **3. Which of the four process activities broke down (or worked well)?**

The major failure occurred during the evolution phase, as the software was stuck on Version 12 from 2004 until 2025, unable to accommodate contemporary needs such as online benefits processing, which led to risky manual workarounds (*The National*, 2014; Auditor-General Report, 2025). The validation process also failed; offline payments bypass automated reasonable checks and independent verification, resulting in significant errors, such as the payment made to the deceased officer going unnoticed until the audit (*The National*, Jan 2026).

### **4. Which attributes of good software were violated or achieved?**

Dependability and security were greatly compromised, as the system was unable to prevent or identify misleading overpayments made to deceased individuals, demonstrating its lack of trustworthiness with public funds. Maintainability was also affected; the outdated architecture obstructed updates, resulting in a dependence on inefficient and error-prone manual processes that reduced the overall system efficiency and stakeholder acceptance.

### **5. Name TWO software engineering practices that would most likely have changed the outcome.**

Ongoing Evolution would have avoided outdatedness; frequent updates would have created up-to-date validation rules to automatically identify issues such as excessive payments. Moreover, Strict Validation Protocols that support and carry out digital approvals and automated cross-checking for all transactions (including manual calculations) would have caught the K439,039 mistake right before the funds were released.

**6. PNG lens: What would an engineer in PNG learn from this case for local delivery conditions?**

Engineers in PNG should focus on creating sustainable maintenance plans while deploying systems. They must ensure there is local capacity to support the long-term development of these systems and avoid returning to insecure manual processes. This case shows that it is important to design with governance in mind. Software should automatically follow local rules, like General Orders, to prevent technical limitations from becoming ways to bypass regulations in resource-limited settings.